

# D1.5: Recurrence Relations

## 1. What Is a Recurrence Relation?

A **recurrence relation** defines a sequence  $u_0, u_1, u_2, \dots$  by expressing each term as a function of earlier terms. The **order** is the largest gap between the current term and the earliest term referenced.

- First-order linear:  $u_{n+1} = au_n + b$ .
- Second-order linear:  $u_{n+2} + pu_{n+1} + qu_n = f(n)$ .
- Homogeneous if  $f(n) = 0$  on the RHS.

## 2. First-Order Linear Recurrence

For  $u_{n+1} = au_n + b$ :

**Case  $a = 1$ :**

$u_{n+1} = u_n + b$  — arithmetic;  $u_n = u_0 + nb$ .

**Case  $a \neq 1$ :**

General solution:

$$u_n = Aa^n + \frac{b}{1-a} \quad (\text{constant particular solution}).$$

Apply the initial condition  $u_0$  to find  $A$ .

**Solve**  $u_{n+1} = 3u_n - 4$  **with**  $u_0 = 5$ .

CF:  $Aa^n = A \cdot 3^n$ . PI: try  $u_n = k$  constant:  $k = 3k - 4 \Rightarrow k = 2$ .

$u_n = A \cdot 3^n + 2$ .  $u_0 = 5 \Rightarrow A = 3$ . So  $u_n = 3 \cdot 3^n + 2 = 3^{n+1} + 2$ .

### 3. Second-Order Linear Homogeneous Recurrence

For  $u_{n+2} + pu_{n+1} + qu_n = 0$ , try  $u_n = m^n$ :

$$m^2 + pm + q = 0 \quad (\text{auxiliary equation}).$$

| Roots                          | General solution                             |
|--------------------------------|----------------------------------------------|
| Distinct real $m_1, m_2$       | $u_n = Am_1^n + Bm_2^n$                      |
| Repeated real $m$              | $u_n = (A + Bn)m^n$                          |
| Complex $m = re^{\pm i\theta}$ | $u_n = r^n(A \cos n\theta + B \sin n\theta)$ |

**Solve the Fibonacci recurrence**  $F_{n+2} = F_{n+1} + F_n$  **with**  $F_0 = 0, F_1 = 1$ .

Auxiliary:  $m^2 - m - 1 = 0 \Rightarrow m = \phi = \frac{1+\sqrt{5}}{2}$  or  $\psi = \frac{1-\sqrt{5}}{2}$ .

$F_n = A\phi^n + B\psi^n$ . ICs:  $A + B = 0, A\phi + B\psi = 1$ .

Solve:  $A = 1/\sqrt{5}, B = -1/\sqrt{5}$ . Binet's formula:

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

### 4. Non-Homogeneous: Particular Solution (PI)

For  $u_{n+2} + pu_{n+1} + qu_n = f(n)$ :

$$u_n = u_{n,CF} + u_{n,PI}.$$

| $f(n)$                   | Trial PI                                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Constant $c$             | $u_n = k$ constant                                                                                  |
| Polynomial of degree $d$ | Polynomial of the same degree                                                                       |
| $ck^n$                   | $Ak^n$ if $k$ is not a root of the aux eqn; $Ank^n$ if it's a simple root; $An^2k^n$ if double root |

**Solve**  $u_{n+2} - 3u_{n+1} + 2u_n = 4$  **with**  $u_0 = 1, u_1 = 2$ .

Aux:  $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$ .  $u_{n,CF} = A + B \cdot 2^n$ .

But 1 is a root (so constant  $k$  alone doesn't work); try  $u_n = kn$ : substitute

$k(n+2) - 3k(n+1) + 2kn = 4 \Rightarrow -k = 4 \Rightarrow k = -4$ . PI:  $-4n$ .

GS:  $u_n = A + B \cdot 2^n - 4n$ . ICs:  $A + B = 1, A + 2B - 4 = 2 \Rightarrow A + 2B = 6$ .

Subtract:  $B = 5, A = -4$ . So  $u_n = -4 + 5 \cdot 2^n - 4n$ .

## 5. Applications

### Compound Interest (First Order)

$B_{n+1} = (1 + r)B_n + D$  (balance after  $n$ th period;  $D$  is any monthly deposit).

### Population with Immigration (First Order)

$P_{n+1} = kP_n + I$ , where  $k$  is the growth rate and  $I$  is a constant influx.

### Tower of Hanoi

$T_n = 2T_{n-1} + 1, T_1 = 1$ , giving  $T_n = 2^n - 1$ .

### Counting Combinatorial Structures

The number of binary strings of length  $n$  without consecutive 1s satisfies  $a_n = a_{n-1} + a_{n-2}$  — Fibonacci-like.

## 6. Behaviour as $n \rightarrow \infty$

- If  $|m_i| < 1$  for all aux roots,  $u_n \rightarrow \text{PI}$  (decaying transient).
- If some  $|m_i| > 1$ ,  $u_n$  grows without bound.
- If  $|m_i| = 1$ , behaviour depends on multiplicity and other roots.

## 7. Exam Style Strategy

- For first-order linear: find CF + PI, apply initial condition.
- For second-order: solve aux, identify CF form, then PI using the right trial.
- Always check with  $u_0$  and  $u_1$  that your formula reproduces them.
- If the trial form clashes with CF, multiply by  $n$  (or  $n^2$  for double clash) — exactly parallels DEs in §1.8.